Statistics Experience

> **!** Important
>
> This assignment is due on **Tuesday, December 1st at 11:59pm**.

The world of statistics and data science is vast and continually growing! The goal of the statistics experience assignments is to help you engage with the statistics and data science communities outside of the classroom.

You may submit the statistics experience assignment anytime between now and the deadline.

Each experience has three parts:

1 Have a statistics experience.

2 Make a slide reflecting on your experience.

3 Complete the *Statistics Experience survey.*

You must complete all parts to receive credit.

Part 1: Experience statistics outside of the classroom

Complete an activity in one of the categories below. Under each category are suggested activities. If you do not see something you'd like to do in Categories 1 - 8, you may do Category 9: Choose your own adventure! If there is an activity you'd like to do but you're not sure if it qualifies for the statistics experience, just ask!

Category 1: Attend a talk or conference

Attend a talk, panel, or conference related to statistics or data science. If you are attending a single talk or panel, it must be *at least 30 minutes* to count towards the statistics experience. The event can be in-person or online. If it's close to 30 minutes, but you are very interested in the talk, please reach out and confirm with me.

Category 2: Talk with a statistician/ data scientist

Talk with someone who uses statistics in their daily work. This could include a professor, professional working in industry, graduate student, etc. This can not be me, although I'm happy to always chat about anything you would like!

Category 3: Listen to a podcast / watch a video

Listen to a podcast or watch a video about statistics and data science. The podcasts / videos must be (1) one podcast or video that is at least 30 minutes or (2) multiple podcasts and/or videos that are at least 30 minutes combined.

A few data science podcasts are below:

- Stats + Stories Podcast
- Causal Inference Podcast
- The Test Set by Posit
- Data Skeptic
- posit::conf talks (formally called rstudio::conf)
  - 2025 conference
  - 2024 conference
  - 2023 conference

This list is not exhaustive. You may listen to other podcasts or watch other statistics/data science videos not included on this list. Ask Professor Meyer if you are unsure whether a particular podcast or video will count towards the statistics experience.

Category 4: Participate in a data science competition or challenge

Participate in a statistics or data science competition. You can participate individually or with a team.

Category 5: Read a book about statistics/data science

There are *a lot* of books about statistics, data science, and related topics. A few suggestions are below. If you decide to read a book that isn't on this list, you can ask Professor Meyer if you're unsure whether it will count for the statistics experience. Many of these books are available through NC State library.

- *Data-Driven Hollywood: The New Data Professionals in the Age of Streaming* by Violaine Roussel

- *Data Empire: The Power of Information to Organize, Control, and Dominate* by Roopika Risam
- *CHART: Designing Creative Data Visualizations from Charts to Art* by Nadieh Bremer
- *W. E. B. Du Bois's Data Portraits: Visualizing Black America* by Britt Rusert and Whitney Battle-Baptiste
- *How Charts Lie: Getting Smarter about Visual Information* by Alberto Cairo
- *The Theory that Would Not Die* by Sharon Bertsch McGrayne
- *The Lady Tasting Tea* by David Salsburg
- List of books about data science ethics

This list is not exhaustive.

Category 6: TidyTuesday

Another option is participating in a TidyTuesday challenge. New data sets are announced on Monday afternoons. You can find more information about TidyTuesday and see the data in the TidyTuesday GitHub repo. This may be a good option if you are familiar with coding!

A few guidelines:

Go to the GitHub link above, and find a 2026 data set to explore.

- Ask an interesting question about the data
- Create data visualization(s) to help address this question
- Summarize where the data come from/how the data are collected, why you found your question interesting, and your descriptive answer to your question.

Include the link to your GitHub repo in the slide summarizing your experience.

Category 7: Coding Out Loud

This may be a good option if you are familiar with coding!

Watch an episode of Coding Out Loud and work through the project. A few guidelines:

- Create a .R, RMarkdown, or Quarto file with all the code needed to reproduce your visualization.

The final product (visualization, table, etc.) should include features or customization that are beyond what is achieved in the Coding Out Loud episode.

Make sure to include why you were interested in this episode, what you learned, etc.

Category 8: CURV - connecting, uplifting, and recognizing voices

CURV is a project by Dr. Jo Hardin at Pomona College to highlight statisticians and data scientists from groups who have been historically marginalized in the discipline.

- Learn about and summarize one of the scholars in the CURV data base
- For this statistics experience, you can contribute to the CURV database. If there is a scholar you would like to suggest for the database, please reach out to me.

Category 9: Choose your own adventure!

Perhaps you're interested in a Statistics Experience doesn't seem to fit in one of the previous categories. If so, you can create your own experience! Please share your idea with Professor Meyer to get it approved for this assignment.

Part 2: Reflect on your experience

Make one slide summarizing and reflecting on your experience. Post an image of the slide in the Statistics experience discussion thread on Canvas.

Include the following on your slide:

- **Description of the experience**
  - Name and brief description of the event/podcast/competition/etc.
- **Something you learned**
  - Write 2 - 4 sentences about something you learned or found particularly interesting or unexpected.
- **Connection to STA 312**
  - Write 2 - 4 sentences about how the experience connects to what we've done in the course.
- **Citation or link to web page for event/competition/etc.**
  - No citation needed if you do an interview.

Make sure the slide includes the information mentioned above and is easily readable (i.e. use a reasonable font size!). **Creativity on the experience and slide design is encouraged!**

Part 3: Complete the Statistics Experience survey

Coming Soon! to complete the Statistics Experience survey. The end of the survey includes a consent asking whether you agree to let your de-identified responses be used for research.

The survey data will be collected and de-identified by our teaching assistant. The survey is optional, and for extra credit. If you do not consent to be part of the research study, you do not need complete the survey. Professor Meyer will receive the de-identified survey responses after the final course grades have been submitted.

If you decide that you do not want to participate in our research study by filling out the survey, you can complete an alternative assignment to earn the same extra credit. In 2-3 hand-written paragraphs, please answer the following questions:

– On your own, provide working definitions of performance assessment, and interest-based assessment. Cite your sources in APA format. Describe one general benefit from each assessment type.

– Next, reflect on your experience completing homework assignments, and how this experience contrasted with your statistics experience assignment. Compare and contrast each assessment based on what you preferred (+ why) and what assessment taught you the most about statistics. Provide one additional contrast you found interesting. This will also be turned in via Gradescope.

Submission

This assignment is due on **Tuesday, December 1st at 11:59pm.** To be considered on time, the image of the slide must be uploaded to Gradescope and the survey completed by the deadline. There is no late policy for the statistics experience.

Completing the survey will earn you extra credit towards the final exam. You may earn the same amount of extra credit by writing a 1-paragraph reflection on your experiences between completing a standard homework assignment vs the statistics experience. You will earn +3 extra credit on either Exam-1, or Exam-2. Note, you can not earn more than a 100% on the exam.

Examples

Please see the slides from day 1 here!

Template

You are highly encouraged to be as creative as possible. I also want you to practice being professional. You can use a NCSU template here or create your own!

Grading

This assignment will be graded out of 50 points based on:

– the quality and depth of your writing/slide

– creativity

– description of the experience

– connection to ST312 (can be a loose connection)